MC-12 technical note: tube and rotor compatibility

Fictional document for Northfield Lab Supply, a fictional company created for a portfolio project. Not for real use.

Item	Value
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Product	Benchtop Microcentrifuge MC-12
Applies to	NLS-MC12-STD, NLS-MC12-PLS
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Purpose

This note explains which tubes and rotors can be used together, and how to balance a load. It applies to both the Standard and the Plus unless stated.

Compatible tubes

- 12-place rotor (both versions): 1.5 ml and 2.0 ml snap-cap tubes, including Northfield microcentrifuge tubes 1.5 ml (rated to 20,000 ×g).
- 8-place PCR strip rotor (Plus only): 0.2 ml tube strips. This rotor is not available for the Standard.
- Tubes must be rated for at least the maximum RCF of your version: 14,200 ×g (Standard) or 18,900 ×g (Plus).

Balancing

1. Load tubes in opposite positions.
2. For an odd number of tubes, add a balance tube of the same type with the same volume of water.
3. Check that liquid volumes in opposite tubes differ by no more than 0.1 ml.

Speed and RCF

RCF depends on speed and rotor radius: RCF = 1.118 × 10^-5 × radius (cm) × rpm². For the 12-place rotor (7.5 cm) this gives 1,300 ×g at 4,000 rpm, 5,400 ×g at 8,000 rpm and 14,200 ×g at 13,000 rpm.

Not covered

Use with tubes larger than 2.0 ml, or with adapters not supplied by Northfield, is not covered by this note.